

FINAL EXAM PORTFOLIO

ITAI 4370: AI 5/6G Communications & O-RAN Networks

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A module-by-module record of course activities, results, technical growth, and reflection.

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Portfolio Overview

Purpose. This portfolio documents the major activities completed in ITAI 4370 and explains what I learned from each experience. The course moved from basic telecommunications concepts into digital signaling, 5G core functions, edge computing, artificial intelligence in the radio access network, Open RAN architecture, responsible AI, and the future direction of 6G.

Organization. The entries are arranged by course topic and completed assignment sequence. Each module contains a summary of the activity, the result or evidence produced, and a reflection explaining how the work increased my understanding of AI-enabled wireless networks.

Visuals and intellectual property. All figures in this portfolio are original visual reconstructions created to document course concepts and results. External standards and frameworks are credited in the References section and cited in the relevant modules.

Module	Main Topic	Primary Evidence
1	Telecommunications Fundamentals	Communication components, media, metrics, services, and topologies
2	Digital Communication and BPSK	Signal flow, modulation, channel noise, and bit recovery
3	5G Core and Edge Computing	EPC versus 5GC, network slicing, MEC, resource allocation, and NRF
4	AI/ML and O-RAN	SON, predictive maintenance, AI resource control, Open RAN, and RIC functions
5	Responsible AI and 6G	Fairness, privacy, security, accountability, governance, and IMT-2030

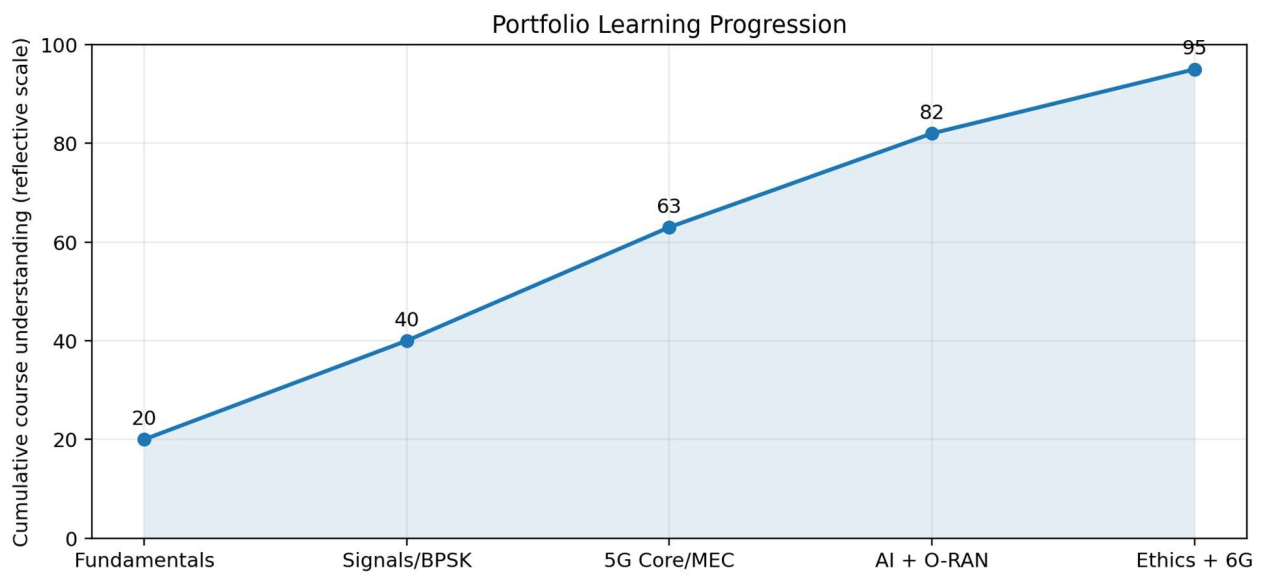


Figure 1. Reflective learning progression across the five portfolio modules.

Module 1: Telecommunications Fundamentals

Activity and Evidence

The first assignment focused on the basic language and structure of telecommunications. I explained the scope of telecommunications, identified the information source, transmitter, channel, receiver, and destination, and compared analog and digital communication. I also reviewed voice, data, video, and multimedia services and compared star, mesh, ring, and bus network topologies.

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
Assignment 01	Answered five telecommunications fundamentals questions using real examples such as FM radio, VoIP, Zoom, and messaging applications.	Connected textbook concepts to systems people use every day.
Component analysis	Mapped each stage of a communication system from information generation to delivery.	Developed a repeatable method for analyzing unfamiliar networks.
Topology comparison	Compared reliability, cost, scalability, and failure points for star, mesh, ring, and bus layouts.	Learned that architecture must match the mission and acceptable level of risk.
Performance metrics	Reviewed bandwidth, latency, throughput, reliability, signal quality, and interference.	Recognized that one metric cannot describe total network performance.



Noise and interference can affect the channel and the received signal.

Figure 2. Basic communication-system model used in the Module 1 analysis.

Results

- I can separate a communication system into functional blocks and explain what each block contributes.
- I can distinguish analog signals from digital data and explain why modern networks increasingly rely on digital processing.
- I can compare network topologies based on redundancy, cost, maintenance, and scalability rather than choosing one design for every situation.
- I developed a stronger foundation for understanding later 5G and O-RAN topics because those advanced networks still depend on the same basic communication chain.

What I learned: The main lesson was that advanced wireless systems are built on basic engineering decisions. Before discussing AI or 6G, I need to understand how information is created, encoded, transmitted, affected by the channel, recovered, and delivered.

Module 2: Digital Communication and BPSK

Activity and Evidence

In the communication-system activity, I created a conceptual signal-flow diagram and examined a Binary Phase Shift Keying (BPSK) transmission. The activity followed source bits through modulation, a noisy channel, receiver detection, and recovered output bits. This made the effect of channel noise easier to see than a definition alone.

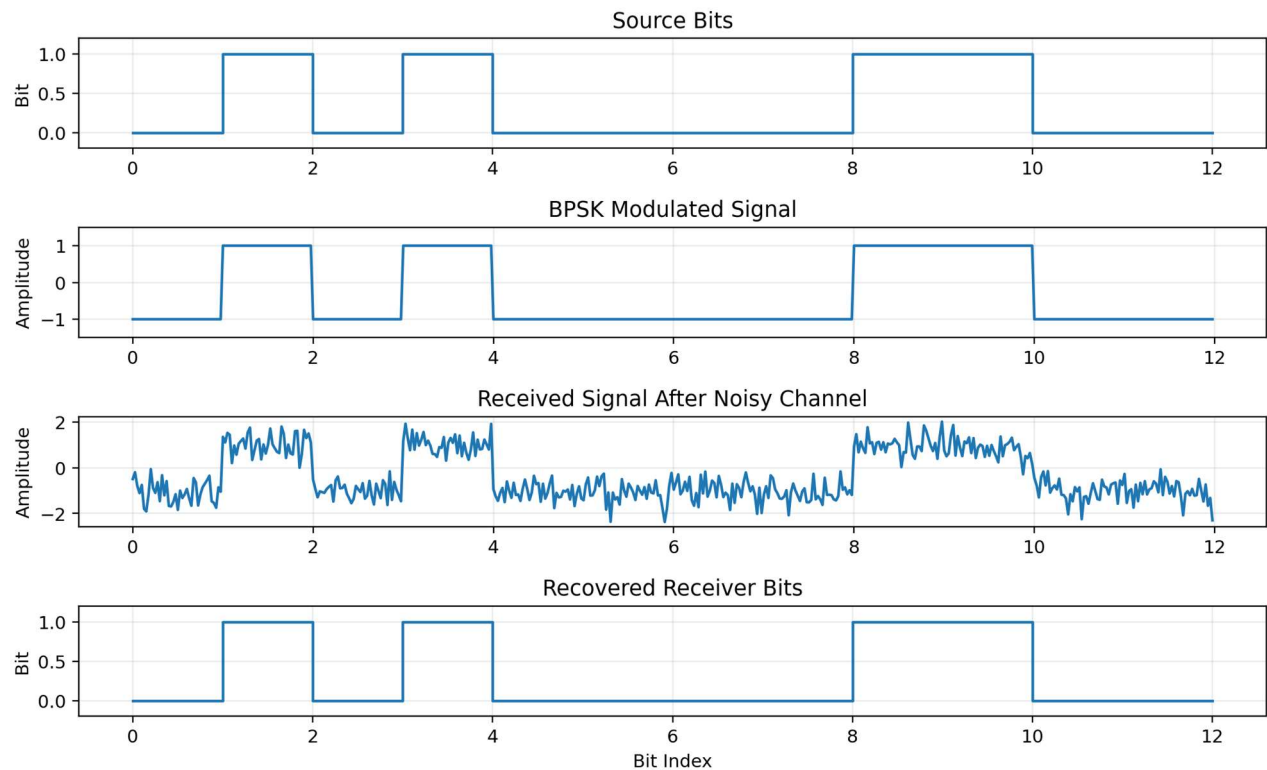


Figure 3. Representative reconstruction of the course BPSK activity: source bits, modulated signal, noisy received signal, and recovered bits.

Process

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
Source data	Started with a short binary sequence represented by zeros and ones.	Established the original message used to check receiver accuracy.
BPSK mapping	Mapped binary values to two opposite signal phases or amplitudes, commonly represented as -1 and +1.	Converted bits into a waveform that can be transmitted.
Noisy channel	Added random variation to represent interference and channel effects.	Observed that the received waveform can look distorted even when the original information remains recoverable.
Receiver decision	Averaged or detected each bit interval and used a threshold to decide whether the bit was a zero or one.	Demonstrated how a receiver converts an imperfect waveform back into digital data.
Verification	Compared recovered bits with the	Used bit-level comparison as a simple

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
	source sequence.	performance check.

Results and Reflection

The representative reconstruction recovered the transmitted sequence under moderate noise. The important result was not that every channel will behave this cleanly, but that modulation and receiver design create a structured way to protect and recover information. When noise increases, the probability of incorrect bit decisions also increases, which connects signal quality directly to reliability.

What I learned: A signal can look messy and still contain usable information. Digital receivers do not need a perfect copy of the transmitted waveform; they need enough separation to make accurate decisions. This activity helped me connect mathematics, plots, and practical network performance.

Module 3: 5G Core, Network Slicing, and MEC

Activity and Evidence

Assignment 02 moved from basic signals into mobile-network architecture. I compared the 4G Evolved Packet Core (EPC) with the 5G Core (5GC), explained network slicing, described Multi-access Edge Computing (MEC), discussed AI-assisted dynamic resource allocation, and explained how the Network Repository Function (NRF) supports service discovery in the 5G service-based architecture.

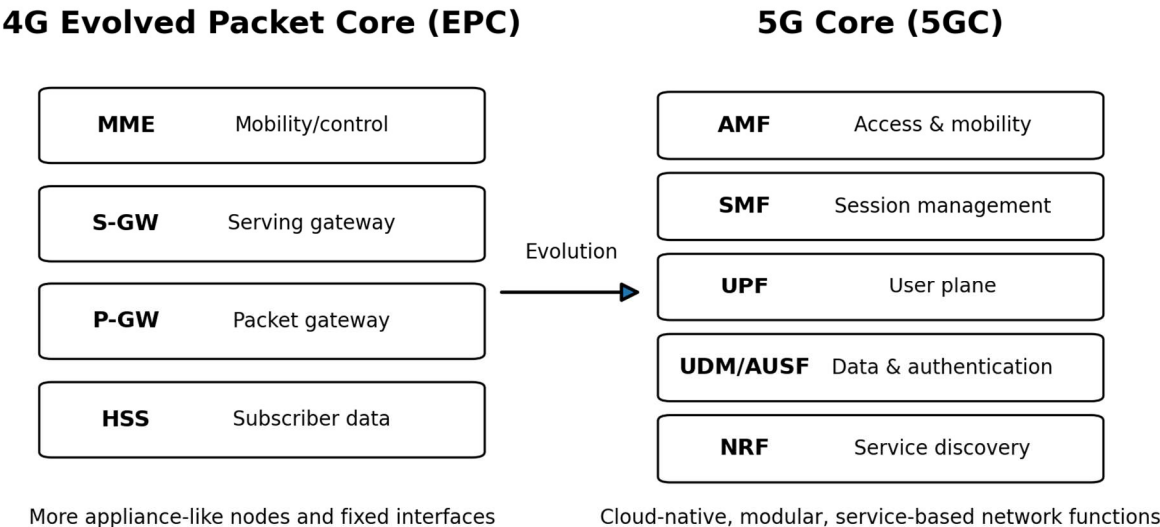


Figure 4. Simplified comparison of major 4G EPC and 5G Core functions.

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
EPC versus 5GC	Compared appliance-style EPC nodes with cloud-native 5G network functions.	Understood why modular network functions improve flexibility and automation.
Network slicing	Designed an EMS-focused example with priority telemetry, video consultation, location, and public-safety traffic.	Learned that one physical network can support different logical service requirements.
MEC	Placed computing resources close to users and devices.	Connected edge location to lower latency, reduced backhaul traffic, and faster decisions.
Dynamic allocation	Used demand and priority values to divide limited capacity.	Saw how AI can continuously adapt resources instead of relying only on fixed rules.
NRF	Explained registration and discovery of 5G network functions.	Understood that service discovery is essential in a modular service-based core.

Applied EMS Example

My EMS background made network slicing and MEC especially practical. A field ambulance may need continuous patient telemetry, a live video connection with a physician, location data, and ordinary

administrative traffic at the same time. Those services should not be treated equally during congestion. A public-safety or medical slice can protect the most time-sensitive traffic, while an edge application can analyze data closer to the incident instead of sending every task to a distant cloud.

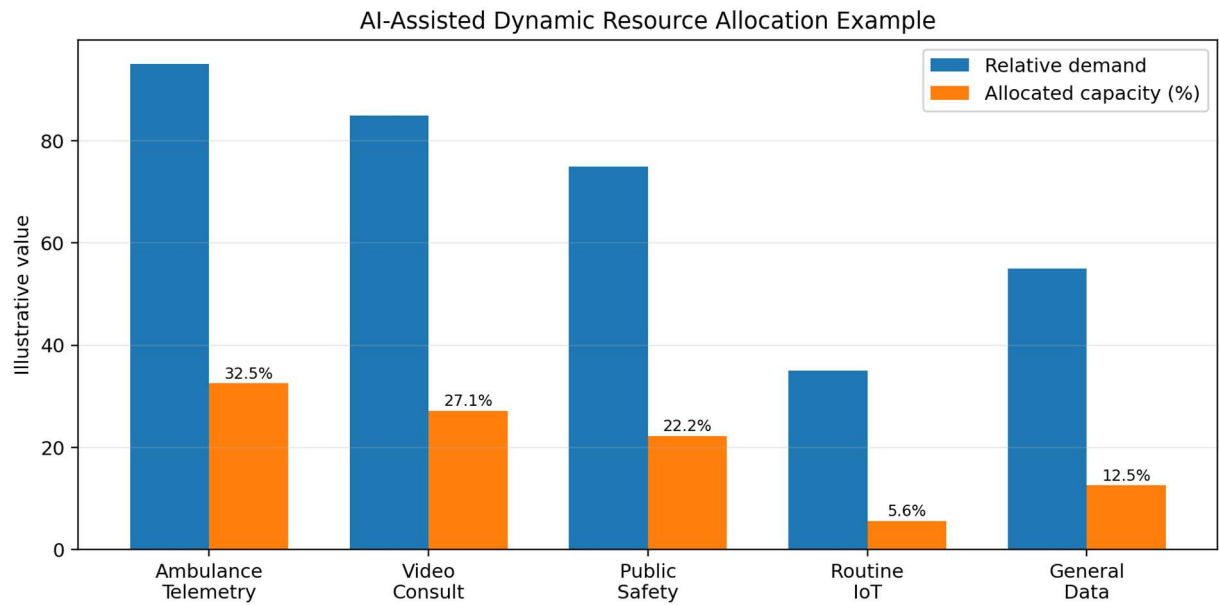


Figure 5. Illustrative AI-assisted capacity allocation that gives greater weight to time-sensitive EMS and public-safety traffic.

What I learned: 5G is more than a faster radio. Its core architecture, slicing, virtualization, service discovery, and edge computing make the network programmable. This module showed me how software and AI can turn network capacity into a flexible service platform. The 5G Core uses a service-based architecture, while network slicing and edge computing support specialized services and faster response times.

Module 4: AI/ML in RAN and O-RAN

Activity and Evidence

Assignment 03 focused on how artificial intelligence and machine learning can improve the Radio Access Network (RAN). I examined self-organizing networks, self-healing, predictive maintenance, and AI-based resource allocation. I also compared traditional RAN designs with Open RAN and explained the roles of the Near-Real-Time RAN Intelligent Controller and the Non-Real-Time RAN Intelligent Controller.

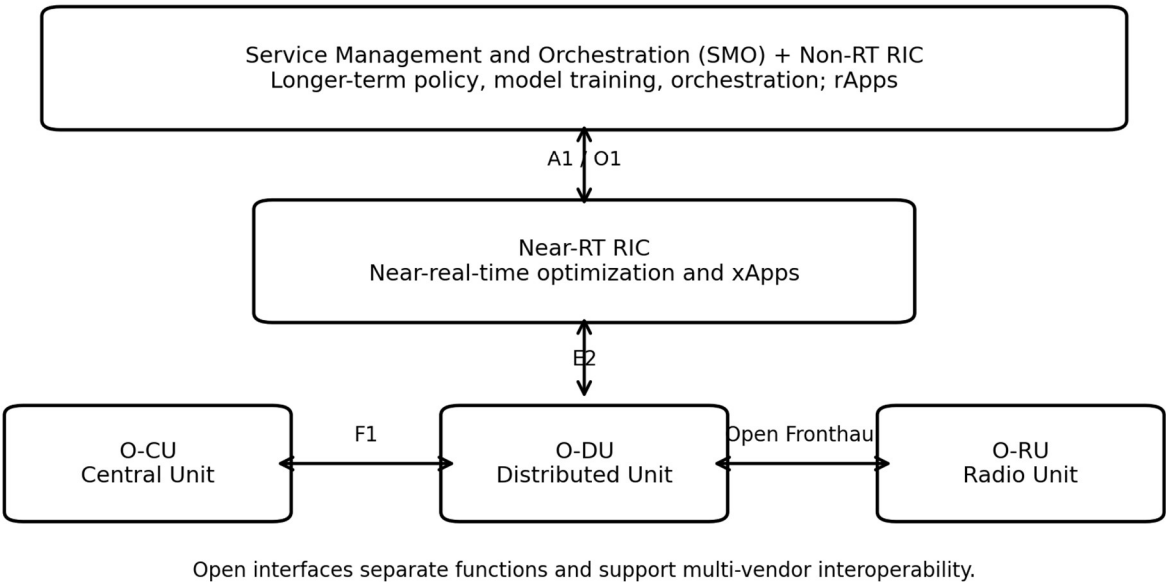


Figure 6. Simplified O-RAN architecture showing the SMO, Non-RT RIC, Near-RT RIC, O-CU, O-DU, and O-RU.

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
AI/ML in RAN	Identified use cases including traffic prediction, handover optimization, anomaly detection, energy saving, and interference management.	Connected machine learning outputs to real network-control decisions.
SON and self-healing	Explained automatic configuration, optimization, fault detection, and recovery.	Learned how automation can reduce downtime and manual troubleshooting.
Predictive maintenance	Used performance history and fault patterns to predict equipment problems before failure.	Understood how data-driven maintenance can improve reliability and cost control.
Python allocation example	Normalized user demand values and converted them into proportional resource shares.	Practiced turning an engineering rule into a simple algorithm.
Traditional RAN vs. O-RAN	Compared vertically integrated vendor systems with disaggregated functions and open interfaces.	Recognized the benefits and integration challenges of multi-vendor interoperability.
RIC timing roles	Separated long-term policy/model work	Understood why different control loops

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
	in the Non-RT RIC from near-real-time control through the Near-RT RIC.	operate at different time scales.

Results and Reflection

The strongest result from this module was understanding where AI fits in the network. AI is not simply placed everywhere. Long-term model training, policy, and orchestration belong in slower control loops, while near-real-time applications use current RAN measurements to optimize resources. Open interfaces make these applications more portable, but openness also creates testing, security, integration, and accountability challenges.

What I learned: O-RAN separates network functions so that intelligence can be added through rApps and xApps. The Non-RT RIC supports longer-term optimization and AI/ML model management, while the Near-RT RIC supports near-real-time control through the E2 interface [5]-[6]. The technology creates more competition and flexibility, but success depends on interoperability and secure implementation.

Module 5: Responsible AI and the 6G Outlook

Activity and Evidence

Assignment 04 examined ethics in artificial intelligence through conceptual, scenario-based, and reflective questions. I reviewed fairness, bias, transparency, explainability, privacy, cybersecurity, accountability, intellectual property, environmental effects, and human oversight. I then connected these issues to AI-controlled telecommunications systems and future 6G networks.

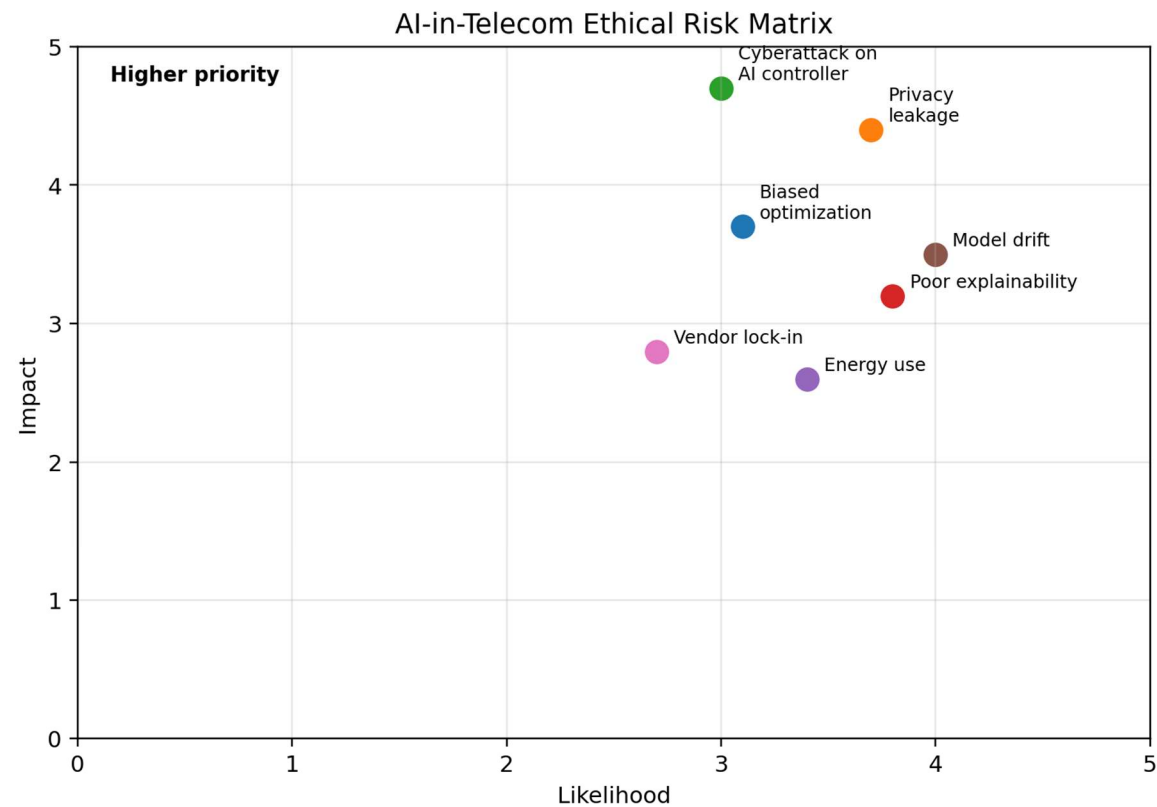


Figure 7. Ethical and operational risks that should be assessed before deploying AI in telecom control systems.

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
Fairness and bias	Considered whether optimization data or objectives could reduce service quality for particular locations or users.	Learned to evaluate who benefits and who may be disadvantaged by an automated decision.
Privacy	Reviewed the sensitivity of location, usage, subscriber, device, and network-behavior data.	Recognized that data minimization and access control are design requirements.
Transparency	Asked how operators can understand, document, and challenge AI recommendations.	Connected explainability to troubleshooting and accountability.
Cybersecurity	Considered attacks on training data, models, control interfaces, and automated policies.	Learned that an intelligent controller can become a high-value attack target.
Human oversight	Kept human responsibility for high-	Understood that automation should support operators, not remove

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
	impact or safety-critical decisions.	responsibility.
Environmental impact	Considered the energy cost of model training, inference, dense infrastructure, and edge clouds.	Connected technical performance to sustainability and lifecycle decisions.

6G Outlook

The course also prepared me to view 6G as an evolution toward more intelligent, integrated, and adaptive networks rather than only another speed increase. The ITU uses the name IMT-2030 for the 6G framework. Expected directions include AI-native networking, integrated sensing and communication, stronger support for immersive services and digital twins, improved sustainability, broader coverage, and tighter communication-computing integration [1]. O-RAN research is also extending RIC concepts toward 6G and deeper AI/ML integration [6].

What I learned: Responsible AI has to be designed into the network from the beginning. NIST organizes AI risk work around Govern, Map, Measure, and Manage, while UNESCO and the OECD emphasize human rights, fairness, transparency, accountability, and trustworthy AI [7]-[9]. These ideas are directly relevant to telecom because automated network decisions can affect access, safety, privacy, and critical services.

Final Course Reflection

At the beginning of ITAI 4370, I mainly thought of telecommunications as moving voice and data from one place to another. By the end of the course, I understood the network as a layered system that combines radio engineering, cloud-native software, edge computing, data, machine learning, automation, cybersecurity, and ethics.

The sequence of assignments made sense because each one added another level. Telecommunications fundamentals gave me the vocabulary. The BPSK activity showed what happens to information at the signal level. The 5G Core assignment showed how services are organized and discovered. The AI and O-RAN assignment showed how the RAN can become programmable and adaptive. The ethics assignment forced me to consider whether a technically successful system is also fair, secure, explainable, and responsible.

The EMS use case was the strongest personal connection for me. In emergency operations, latency, reliability, prioritization, and security are not abstract measurements. They can affect the quality and timing of care. Network slicing, MEC, AI-assisted resource control, and resilient O-RAN systems could support real-time telemetry, video consultation, location services, and public-safety communications. At the same time, automated decisions must be tested carefully because failures or biased policies could affect people during critical moments.

This course also connected directly to my AI and robotics degree. Future robots, autonomous vehicles, digital twins, smart medical systems, and industrial devices will depend on dependable communication. AI models cannot act on current conditions if the network cannot deliver accurate data with the required speed and reliability. I now see 5G, O-RAN, and future 6G systems as part of the intelligence pipeline, not just the connection between devices.

Skills I Can Carry Forward

- Analyze a communication system from source to destination.
- Interpret digital modulation and noisy-channel plots at a basic level.
- Explain major 5G Core functions, service-based architecture, network slicing, MEC, and NRF service discovery.
- Describe AI/ML use cases for RAN optimization, self-healing, predictive maintenance, and resource allocation.
- Compare traditional RAN and O-RAN and explain Near-RT and Non-RT RIC responsibilities.
- Identify ethical, privacy, security, transparency, sustainability, and human-oversight requirements for AI-enabled networks.
- Connect wireless-network concepts to EMS, robotics, digital twins, industrial IoT, and future autonomous systems.

Portfolio Artifact Inventory

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
Assignment 01	Telecommunications Fundamentals	Five-question written analysis and communication-system figure.
Communication Activity	Conceptual Communication System Diagram and Signal Flow Analysis	Signal-flow diagram and representative BPSK plots.
Assignment 02	5G Core, Network Slicing, MEC, AI Allocation, and NRF	Architecture comparison, EMS application, and allocation chart.
Assignment 03	AI/ML in RAN and O-RAN	O-RAN diagram, AI use cases, RIC

Portfolio Element	Activity / Evidence	Result or Skill Demonstrated
		comparison, and algorithm reflection.
Assignment 04	Ethics in Artificial Intelligence	Conceptual, applied, and reflective ethics analysis plus risk matrix.
Final Portfolio	Integrated course reflection	Module results, learning progression, figures, tables, and cited sources.

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